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Responsible use of Generative AI in chemical engineering

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ABSTRACT

Generative Artificial Intelligence is a rapidly developing area being used to create powerful tools which have the potential to change a wide range of professional practices in chemical engineering. As this area develops, new principles on responsible use of Generative AI in chemical engineering are required to ensure that traditional engineering ethics are able to accommodate the new landscape. In this perspective, we assess the current state of engineering ethics, responsible AI principles and suggest how they can combine to ensure that Generative AI can be used responsibly within the chemical engineering sector. Whilst there are many aspect to engineering ethics and responsible AI use, the core principles which include transparency, integrity, and accountability are omnipresent and provide a shared foundation of good practice on which new regulations may be built as the need arises. Future breakthrough will require development on the AI technology itself, the people-centre approach and regulation changes.

Introduction

Generative artificial intelligence (AI) is a class of AI which is able to create new information after being trained on a set of data, typically text and images. Generative AI holds promise to revolutionise the way we work and learn, through the automation of a variety of tasks to allow complex processes to be completed in short order. Chatbots such as OpenAI's ChatGPT, as an example of Generative AI are large language models (LLMs), have already begun to integrate into daily life. Apart from the commonly seen LLMs, there are also other types of Generative AI for a variety of purposes such as video, sound, image, and 3D model creation.

Chemical engineering, as a discipline, has seen a shift in recent years from a mindset viewing chemicals as low cost and inexhaustible feedstocks to one where efficiency and sustainability are key. The chemical industry is an extreme example of both energy use and greenhouse gas emissions, which makes it uniquely positioned to adopt emerging technologies to help limit and reduce environmental harm (Liao et al., 2022). The new focus can be seen as extensions of two of the core tenets of chemical engineering, safety, and profitability. In an environment where energy prices are rising and resources are dwindling, focus on using less to do more and reusing as much as is possible is simply a prudent way to improve plant economics. The reduced energy usage and material wastage has the incidental advantage of being a socially responsible way of conducting engineering. Safety culture has long been aimed at putting people before profits, and environmental consciousness is a long-term application of this philosophy to future generations. With the fourth industrial revolution (industry 4.0) already underway, the use of digital technology and artificial intelligence to achieve these aims has been increasingly popular. Advanced automation allows system and process design to be optimised with much greater efficiency and accuracy.

Within this context, generative AI has begun to show promise in the streamlining processes within the world of chemical engineering, with uses being found in interactions between systems with complex data, improving warning systems, answering technical questions, and generating computer code (He et al., 2020; Savage, 2023; de Kok, 2023). These examples show Generative AI in its infancy within the chemical engineering sector, and uses such as P&ID generation, HAZOP completion, lifecycle analyses (LCAs), and even complete chemical plant design being envisioned as mature uses (Liao et al., 2022; Hirtreiter et al., 2022).

Instead of allowing a trial and error approach to guide the use of generative AI in chemical engineering, a proactive approach is required to guide its development to maximise the potential benefits of widespread uptake. As a sector, there is a need to expand the strong culture of the ethical consideration of safety and environmental factors to the use of generative AI for practical purposes. Due to the need to consider such a broad range of potential consequences and hazards associated with the chemical industry, such as the immediate dangers and long lasting

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consequences on the environment of chemical production, chemical engineers are strongly positioned to consider a broad range of ethical factors which are applicable to Generative AI use in industry.

Responsible use of Generative AI in chemical engineering

Engineering ethics are an explicit part of being a professional engineer and rely on the acknowledgement of an engineer's potential to affect society both positively and negatively (Royal Academy of Engineering, 2011). Interdisciplinary engineering ethics are available, and the Royal Academy of Engineering states that engineers should act, with amongst others, honesty, respect for life, public good and the environment, and with integrity (STATEMENT OF ETHICAL PRINCIPLES for the engineering profession, 2022). Without the consideration of these ethics within chemical engineering, serious disasters may be caused which are unique to the discipline. Respect for life, law, the environment, and public good are principles deeply rooted in safety culture.

Chemical engineers have a strong background and understanding of both safety and environmental issues due to the unique concerns of the chemical industry. These two issues are closely interlinked with the use of Generative AI in the chemical industry, however chemical engineers are also strongly positioned to deal with more general Generative AI issues as they pertain to the chemical industry, as shown in Fig. 1. Societal issues, whilst not unique to chemical engineering, should be strongly considered due to the danger which chemicals can create. Whilst LLMs do have inbuilt safeguards, the information on chemicals and chemical processes put into these models can be potentially dangerous, particularly if this information could be used to create dangerous chemical weapons.

Safety culture was born from engineering disasters which ranged from minor injuries to major loss of life and environmental damage. Balhorn et al. (2023), discussed the ethical implications of using ChatGPT to answer science and engineering questions for practitioners such as chemical engineers. It was found that ChatGPT has some initial built-in safeguards to prevent unethical use, as it refused an answer to a question on how to synthesise illegal drugs (Balhorn et al., 2023). This use demonstrates the promise of responsible Generative AI; however, we argue that this does not demonstrate inherent safety in the model, which is considered key in chemical engineering design, but rather a limited and reactionary response to individual instances of safety issues in the model which fails to address the key failures of Generative AI (Kletz, 1985). Systemic approaches to generating responses which breach these safeguards have been found which demonstrates the need for future implementations to consider an inherently safer approach to the development of LLMs (Deng et al., 2023). As chemical engineers are responsible for the complete safety structure of a chemical process from design to implementation, with implications for both personnel and the environment, and LLM design for the chemical industry needs to be created with the same strict standards.

Other issues will be created through further widespread use of AI in the chemical industry. There is ongoing research into the digitisation of engineering flow sheets for the purpose of using AI to generate engineering diagrams automatically (Elyan et al., 2020). If any form of chemical engineering diagram is used without adequate assessment, the results could be disastrous.

The fundamentals of responsible Generative AI use in chemical engineering should focus on the potential impacts to individuals and wider society, to prevent its use from causing more harm than good due to unintended side effects. The principle of considering the effects of a project on individuals and the environment has already been widely adopted in chemical engineering. HAZOPs and LCAs are designed to improve the health and safety of both plant operators, the surrounding areas, and future generations. However, as AI becomes ubiquitous, these principles need developing for the emerging area of AI use in chemical engineering. Furthermore, in order to integrate Generative AI in chemical engineering usage, responsible and safe data collection and usage is required, both from implementation and stewardship perspectives in order to construct the complex datasets required for training models (He et al., 2020; Rane, 2023). Whilst we acknowledge that these challenges are not unique to the chemical engineering sector, chemical engineers are able to leverage their unique skillsets to ensure these challenges are solved responsibly to maximise the benefits of Generative AI. It should also be noted that AI models are complex and often treated as black-boxes, and so an important point of the deployment is the transparency and explainability of model development and results.

An area with a great deal of promise for the use of generative AI in

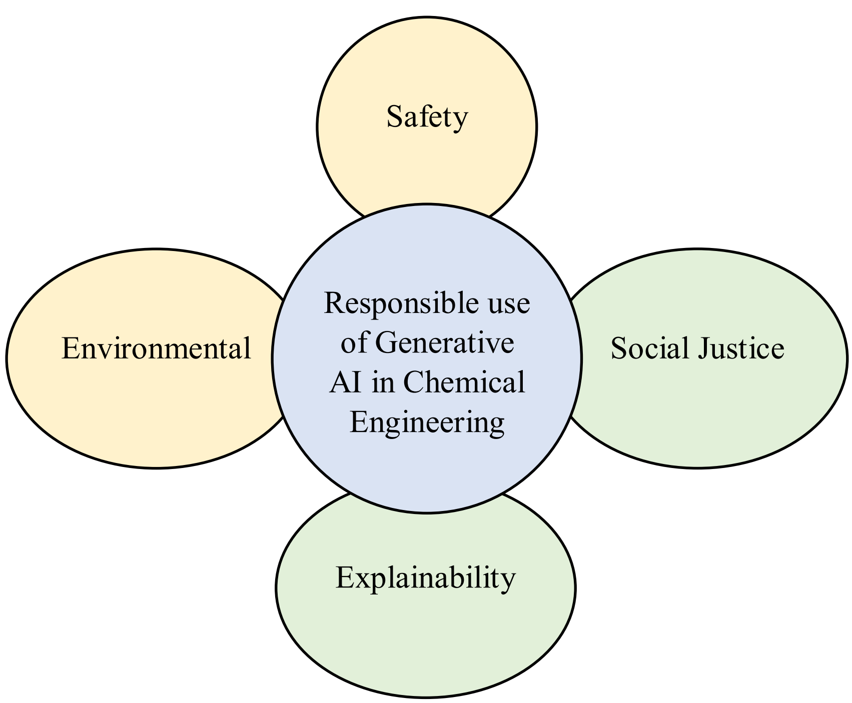

Fig. 1. Core aspects of responsible use of Generative AI highly applicable toChemical Engineering(yellow) and more generally applicable(green).

chemical engineering is that of flow sheeting and process and instrumentation diagrams (P&IDs). Generative AI can theoretically be used to develop flowsheets and P&IDs automatically, based on chemical process information and engineering recommendations, dramatically reducing the time engineers spend drawing and redrawing diagrams. This innovations could increase the speed at which designs are optimised and safety analyses occur, however at present several hinderances remain. A significant problem is creating databases of flow sheets, which are often not made publicly available, or which may not be in a machine readable format (Hirtreiter et al., 2022). Generative AI typically requires large training datasets for useful models to be created and without a large central repository of flowsheets to learn from models may be limited. Additionally, current attempts at flowsheet automation have not been well integrated with physical knowledge, and so often generate functionally meaningless flowsheet suggestions (Vogel et al., 2023). Without a responsible approach to the use of AI in this scenario, flowsheets which have errors, redundancies, or useless features may slip past review. This highlights the need to retain knowledge and human involvement even when automation is at an advanced level. An irresponsible use of this AI may lead to poorly understood designs being rushed into development which then creates a fundamental safety hazard. To help aid with the problems of flowchart generation lacking large databases to learn from, we believe new standards should arise to ensure that all flowcharts are kept in machine-readable formats.

Development of new materials and structures has been bolstered by new Generative AI methods recently, with electrode design improvements showing particular promise with advanced AI methods (Niu et al., 2023). Integration of physics-based and data driven models presents unique opportunities for design progression and the responsible use of Generative AI. The use of Generative AI promises to speed the development of new electrochemical technologies, such as fuel cells and batteries (Leonard et al., 2021). These technologies are typically needed for the implementation of green systems or decarbonisation of industrial sectors, and so the principles of responsible AI and application of chemical engineering ethics can be applied towards environmental concern.

These examples of Generative AI use in chemical engineering show a trend toward the integration and use of AI over time and in more and more situations. Here, we envisage that for chemical engineers to adopt responsible Generative AI will require a multifaceted approach that unifies technology, people, and regulations. Technological changes to improve the use of Generative AI will focus on improving the models themselves, reducing bias in datasets, increasing clarity in the source of results, and delivering results that create positive impacts – something which many chemical engineers achieve through their work already (IChemE, 2022). People centred approaches for responsible AI use will focus on the needs of the people using AI and what changes need to be made to accommodate them, to make certain that people are able to use Generative AI safely and responsibly. Finally, regulatory changes need to be made to enshrine the changes necessary for responsible AI to become the standard. By treating the necessary changes for Generative AI to be developed responsibly, chemical engineers will be in a position to shape the framework which is needed to maintain the rigorous standards of any new technology relevant to the chemical sector.

Technology enabled solutions

Adhering to the principles of responsible AI use will lead to the attainment of trustworthy Generative AI, defined by the EU as lawful, ethical, and robust, which is an important goal in the development of AI in chemical engineering (IChemE, 2023). By adhering to these standards, AI can foster trust in its users, which is a key step to wider acceptance and uptake. Illegal AI use, for example by breaking data privacy laws, would prevent the public from trusting that their data is safe in other AI applications, and even if the use weren't illegal but were simply unethical the reaction may be largely the same. This is similar to the loss of public trust caused by chemical accidents which result in injury or environmental damage, and due to the potential benefits of Generative AI for chemical engineers illustrates the importance of responsible use (Gamero et al., 2011).

Building better, more trustworthy Generative AI models will result in better use and integration within chemical engineering, which in turn will lead to the creation of greater quantities of high quality data and allow for the development of more sophisticated functionalities. For trustworthy AI to be built, verification of results is key as it allows the model to be explainable and so build a chain of trust from input to output (Wing, 2021). Thiebes et al. (2021) introduced their principles of trustworthy AI and created a framework for trustworthy AI design, which allowed them to determine useful future avenues of research based on how closely related to the principles the research topic is. Chemical engineering and related areas typically rely on high levels of expert knowledge, the inherent trust that someone who has worked in a subject is able to interpret results which would be meaningless to the layman (Feng et al., 2020). This trust is founded on the ability of the expert to explain the process and how the results were obtained, something which is more complex with the use of AI models, as AI models often operate as black-boxes (Rai, 2020). By combining the expert knowledge of an experienced engineer with machine learning models, the expertise and abilities of the engineer can be augmented, and their decision-making ability given increased speed and accuracy (Schweidtmann et al., 2021). We believe that Generative AI can used to improve the engineering workflow by augmenting the skills which chemical engineers already possess, automating the simple but time consuming tasks and creating more time to focus on more complex matters. This would allow both the volume and quality of work to increase; however, without the results being explainable and hence understandable, there is reduced ability to build trust in the model results which would negate the improvements gained through increased work speed.

Explainable AI (XAI) is the concept of ensuring a clear and understandable link between inputs and results, allowing people to better understand how an AI model generated a result. Ali et al. (2023) introduced metrics to determine data explainability, which is a core principle of trustworthy AI. Lundberg et al. (2020) developed several methods for analysing several popular machine learning methods, to improve their explainability for sectors where decisions made using AI directly affect people and there are legal requirements for explaining why decisions were made. Explainable AI is an important part towards creating trustworthy AI, however aspects such as accountability are also required. Chemical engineers need to develop an understanding of Generative AI tools and how they work in order to ensure the accountability and transparency of the results allows them to be trustworthy.

People centred approaches

A people centred approach to responsible Generative AI use is the acknowledgement that systems need to be designed best for the people who use them in order to provide a synergy between human and machine (Rožanec et al., 2022; IChemE, 2023). People centred approaches to Generative AI involve bearing the needs of those affected by the system in mind from its conception, to ensure all aspects of the design and related systems best serves the person rather than the system. The chemical industry is rife with examples of processes and procedures being designed in a way to work with the people who will be performing the tasks, from manual handling guidance and lone working protocols to continual safety reviews. The experience of designing process around people to best utilise their abilities in a safe way must be applied to new Generative AI tools in order to foster responsible use by chemical engineers.

Serious, unintended consequences can be avoided by considering the effects on society and individuals that irresponsible Generative AI use could have. In the same way that safety features and process revisions would not be accepted without qualified scrutiny, AI assisted safety features and process revisions will still need to be thoroughly reviewed. Recent advances in AI assisted HAZOPs have shown promise and show that the possibility of Industry 4.0 HAZOPs is close to being realised. Ekramipooya et al. (2023) created a tool to improve the quality and speed of HAZOP sessions and Marhavilas et al. (2020) developed an expanded framework for automated risk analysis for HAZOPs. These tools have the potential to speed and improve chemical engineers' work, but if poor oversite allows a disaster to occur, apart from the obvious loss of life and financial consequences, wider implementation of Generative AI tools may be halted, causing stagnation of Generative AI under in the chemical sector, or allowing competitors to gain an advantage. Strong leadership and communication will be essential to ensure a culture that treats emerging AI responsibility as seriously as it treats the ethics that underpin safety culture. Companies must adopt AI principles in the same way that other corporate principles including safety, accountability, and transparency are adopted, as the responsible use of AI lies at the intersection of a variety of principles.

Generative AI shows promise in the area of supporting learning and problem solving ability. Programming and wider software proficiency is increasingly required within chemical engineering; However, the skills themselves are often less important than the way they are used. Used correctly, LLMs have been shown to bolster critical thinking and problem solving and allowed a greater integration of programming skills into traditional style tutorial questions (Tsai et al., 2023). However, the dangers of the use of LLMs for cheating and avoiding rigorous work are apparent, and in order to combat them a fundamental shift in the way assessment occurs may be required (Nikolic et al., 2023). Here we can see that irresponsible use of AI comes both from the user of an LLM to cheat an exam, and from the examiner who does not update examination methods to inhibit LLM based cheating. While it is important to instil an awareness of the consequences of using AI in a prohibited way for assessments, it is equally important to be vigilant of the ways that cheating can occur and to focus on reducing it. Moving forward it will be important to learn to use AI with a high level of rigour and responsibility as many aspects of chemical engineering will be integrated with it.

AI technology is at the forefront of innovation, but also the intersection of established responsible practices, creating a grey area of uncertainty which is the responsibility of chemical engineers to map. The human aspect here is equally important as the others, and just as safety culture requires training and incremental skill development, so too will fostering an environment of responsible Generative AI use in chemical engineering. To achieve a safe chemical engineering environment requires honesty, integrity, accuracy, and rigour so that no shortcuts are taken, fewer mistakes are made, and no assumptions of quality or competence lead to avoidable errors taking place. Ensuring that these principles are adhered to requires their application to cascade from good leadership who clearly communicate the expectations and set a positive example.

Regulatory response

Ensuring that Generative AI is developed and deployed responsibly will require a strong regulatory body and policy implementation. A lack of enforcement and standardisation may result in AI being developed without adhering to the principles of responsible AI. In much the same way that safety standards in engineering have been developed and are upheld by law and regulations, AI usage will be regulated, and it behoves the engineering community to be proactive rather than wait for disaster to necessitate policy implementation. In the past, large chemical engineering related disasters have directly led to the creation of laws designed to prevent repeats and improve general safety; However, we argue that if AI laws developed in the same way, trust in the new technology would be broken and AI development may halt. In the past, the chemical industry has been responsible for various incidents caused by a lack of attention to safety needs, which have not only been tragic and damaging for the companies involved, but also extremely expensive (Kletz, 1985). To guard the investment required to bring industry 4.0 to fruition, careful development of new regulations will be required, which chemical engineers will be in a unique position to help develop due to the continuing association improving safety regulations.

AI policy being developed focuses on the accuracy of the datasets and privacy concerns which, within the context of engineering, means it is likely that companies will develop their own extensive datasets to train models on (Liang et al., 2022). However, EU regulations advocate for the use of database registries which are stewarded by a single organisation but accessible to others (Janssen et al., 2020). These sorts of projects would be understandably expensive and time consuming to administer, and so may only be feasible for large organisations and so incentives to implement them may be required.

We agree with proposals that, in order to ensure database integrity, groups of organisations should collaborate to create shared resources whose storage and administration is distributed evenly (Janssen et al., 2020; Gu and Guo, 2024). However, the complexities of data access and information rights may make the projects seem infeasible at first, creating a situation where regulations are required in order to compel the sharing of data.

The regulation of AI faces the problem of determining the floor and ceiling of data sharing for model creation. Enough needs to happen to compel and incentivise the release of data for reliable datasets to be compiled, and limits need to be placed to prevent abuse and maintain integrity of datasets. This data integrity is key to responsible AI use, as without it the AI models' results could not be explained and would not be trustworthy.

The various professional chemical engineering bodies have codes of professional conduct which all members must abide by, typically including themes such as accountability, integrity, honesty, and transparency (IChemE, 2020). This matches well with the principles of responsible AI, where accountability and transparency are key. Still, responsible AI has several more specific principles such as bias reduction. In an era of Generative AI, the professional ethics already established for chemical engineers work well as a basis for good practice but need further refinement and development. Areas where improvements are required include attribution, where Generative AI are trained using data which is not attributable. The lack of attribution can cause concern for organisations as it may needed in some areas of the professional ethics framework, for example, the definition of Plagiarism and collusion in an era of generative AI. Rather than attempt to start over or adapt the current regulations, adding to the exiting set seems most beneficial as it offers the option to combine tried and tested ethical principles with newly developed ones as the need arises.

Conclusion

Due to the extreme environmental impact of the chemical industry, chemical engineers have a responsibility and a strong incentive to use emerging technology to improve processes and products. However, the use of new Generative AI tools is complicated due to the speed of its development and poorly defined usage limits, which could create safety issues unique to the chemical industry. Responsible Generative AI use is a newly emerging subject area, particularly within the domain of use in chemical engineering. As the use of AI becomes more widespread and the technology develops, consensus over the principles of responsible use is emerging, but more effort is needed to enshrine these principles ensure their adherence. The reduced clarity of the general principles of responsible AI use makes defining their relevance to chemical engineering more difficult but due to the speed which AI technology is developing, it is imperative not to delay adapting them to the needs of chemical engineers. Key points to include are transparency, safety, and trustworthiness, to ensure a clear understanding of results and how they were determined, that AI models are developed to the same high standard present in other safety culture processes, and that people can trust results to be equitable and reliable. Strong governance and leadership are required to disseminate these principles, firstly to individual companies and then fostering a strong.

In the future, and as Generative AI technology becomes more widely used, it is important that the principles of responsible AI use are adopted in a similar way to those which govern safety culture, but without the need for disaster to drive development. The recognition that trustworthy and socially conscious Generative AI is inherently more valuable is important and provides and incentive for organisations to develop this way naturally, but regulations ensuring this trend are also important. By using the principles of responsible AI to create people-centred systems, chemical engineers can begin to utilise Generative AI to augment and improve their work.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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